

NOTE

These size limits refer to the length of the QR code for a single product. In case of [concatenation](#), the total size of the concatenated QR code can exceed this size, but still needs to fit within a valid combination of Version and Quality for the total length of characters needed.

For QR codes, [ISO/IEC 18004:2015](#) currently allows for a maximum payload size of 4296 characters using Level 40 and L ECC Level encoding.

For NFC Tags, the actual maximum size is implementation dependent.

ECC Level

The QR code SHOULD employ level M or higher ECC.

NOTE

A higher level ECC does not help against typical 'reading' issues like shiny surfaces, bad contrast or issues with camera resolution/focus, and lack of camera-app processing dedicated for QR codes. Therefore, in certain situations ECC=L MAY be used as well (e.g. to prevent having to move to a higher Version to fit the payload).

5.1.4. Manual Pairing Code

This section describes the content and format of the Manual Pairing Code, which can be used in certain situations next to, or instead of, the QR code or NFC Tag described above.

5.1.4.1. Content

Payload

The payload of the Manual Pairing Code consists of the following required and optional data elements.

Table 60. Manual Pairing Code Elements

Element	Size (bits)	Required	Notes
VERSION	1	Yes	Shall be 0 Version is encoded as part of first digit of the Manual Pairing Code. A value of 1 is reserved for future extension of the specification.
VID_PID_PRESENT	1	Yes	0: no Vendor ID and Product ID present in Manual Pairing Code 1: Vendor ID and Product ID data included
DISCRIMINATOR	4	Yes	4 Most-significant bits of the 12-bits Discriminator described above
PASSCODE	27	Yes	Same as 27-bit Passcode described above